DEEPAK KATUKURI

AI / ML ENGINEER · COMPUTER VISION · NLP & DEEP LEARNING



CONTACT

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TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, Core Java, C++

Deep Learning: PyTorch, TensorFlow, CNNs, EfficientNet, fine-tuning, timm, Hugging Face

Computer Vision: image classification, medical imaging, CLIP ViT, transfer learning

NLP & RAG: Retrieval-Augmented Generation, DistilBERT, sentence-transformers, contrastive learning, embeddings, vector search, Ollama, prompt engineering

Machine Learning & Stats: Scikit-Learn, classification, ensembles, applied statistics, NumPy, Pandas, Matplotlib, Seaborn

MLOps & Cloud: Docker, Kubernetes, AWS, CI/CD, model deployment, FastAPI

Tools: Git/GitHub, pytest, Jupyter, TensorBoard, IBM Watson

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech – Computer Science & Eng.
MLRIT, India
Jun 2018 – Jul 2022

LANGUAGES

English – Fluent

German – Beginner (A1/A2)

CERTIFICATIONS

PL/SQL – Database Management

Core Java – Software Development

OneShield – P&C Insurance

PROFILE

AI / ML Engineer and Data Science MSc candidate with a computer-science and applied-statistics foundation, focused on deep learning across computer vision and NLP. Built a medical-imaging CNN (0.98 ROC-AUC CT-scan detection), a CLIP multimodal retrieval system, and a framework-free RAG pipeline from scratch — research-oriented, mechanics-explicit engineering — backed by 2.5 years of production software engineering. Seeking an English-speaking machine-learning working-student role in Berlin (15–20h/week).

PROJECTS

Intracranial Hemorrhage Detection – Medical Imaging CNN (Computer Vision)

PyTorch · timm · EfficientNetB0 · RSNA CT dataset

- Achieved 0.98 ROC-AUC and 100% recall on intracranial-hemorrhage binary classification in CT scans by fine-tuning a pretrained EfficientNetB0 CNN on the RSNA dataset with PyTorch and timm.
- Packaged the model into a reusable pipeline with pytest unit tests and CLI configuration for repeatable training and inference.

Food Image-to-Recipe Retrieval – Multimodal Deep Learning (Vision + NLP)

PyTorch · CLIP ViT-B/32 · DistilBERT · InfoNCE · MSc final project

- Built a dual-encoder model mapping food images (frozen CLIP ViT-B/32) and recipe text (DistilBERT, separate ingredient/instruction streams) into a shared 1024-d space trained with symmetric InfoNCE contrastive loss.
- Reached a validation median rank of 10/200 with semantically correct top-5 results on a 13.5K-image Kaggle dataset, and cut training time ~10x by pre-computing and caching CLIP image features.
- Shipped a reproducible, config-driven pipeline (OmegaConf/YAML, TensorBoard, mixed-precision, early stopping) with 58 unit tests and R@k / median-rank evaluation in both retrieval directions.

RAG Pipeline – Retrieval-Augmented Generation from Scratch (NLP)

Python · sentence-transformers · NumPy · Ollama · github.com/Deepak589/RAG_pipeline

- Built a framework-free RAG pipeline in ~270 lines of Python (no LangChain, LlamaIndex, or vector DB), implementing chunking, embedding, similarity search, and prompt assembly to keep every retrieval mechanic explicit.
- Improved retrieval relevance by pairing a dense bi-encoder retriever (all-MiniLM-L6-v2, 384-dim, cosine top-3) with a hand-rolled TF-IDF baseline, plus heading-prefix chunking and Table-of-Contents filtering to cut off-topic matches.

EXPERIENCE

Software Engineer
Mar 2022 – Sep 2024
ValueMomentum

- Maintained 2 enterprise P&C insurance and surety-bond applications on OneShield PAS, resolving production defects across policy workflows.
- Built and optimized PL/SQL stored procedures and functions powering backend insurance data operations and endorsement modules.
- Accelerated releases by running QA/UAT test cases and supporting CI/CD deployment of fixes across the full policy lifecycle.

ML Intern
Jun – Jul 2019
SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it via IBM Watson ML, communicating results with Matplotlib/Seaborn visualizations.